Session 2.7 : Decoders and Encoders

Session 2.7: Focus

- Decoder
  - Basic Binary Decoding logic
- Different n-to-m Binary Decoders
  - 2-to-4 Decoder
  - 3-to-8 line Decoder Implementation
  - 4-bit Decoder
  - Logic Symbols of Decoders
  - Decoder in Use
- Encoder
  - Decimal-to-BCD Encoder

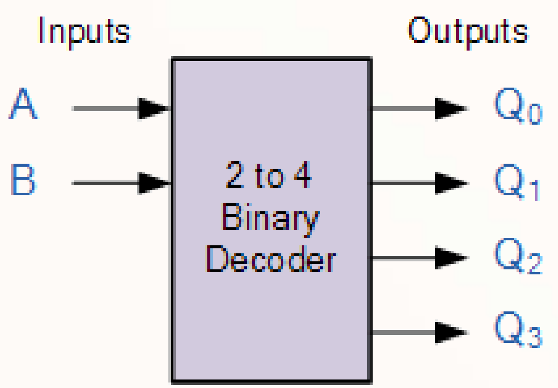
Decoder

Decoder

- A **decoder** is a digital circuit that detects the **presence** of a specified **combinations of bits** (**code**) on its **inputs** and
  - Indicates the **presence** of that **code** by a **specified output** level
- In its general form, a decoder has **n input** lines to handle **n** bits
  - It has from **one** to **2^n output** lines to indicate the **presence** of **one** or **more n-bit combinations**

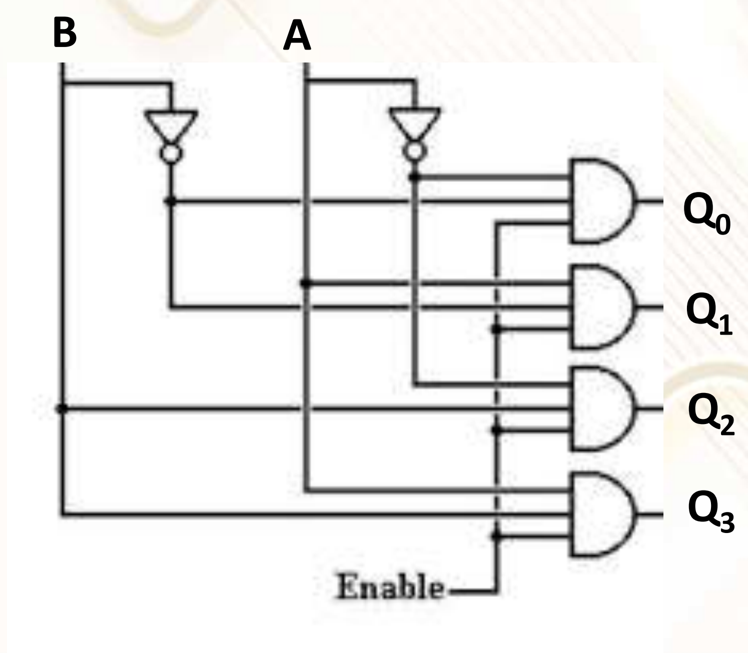
Different n-to-m Binary Decoders

2-to-4 Binary Decoder



Truth Table

B	A	Q_0	Q_1	Q_2	Q_3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1



Note: **Enable** signal if **LOW**,
all outputs are always LOW.
Inputs do not affect the outputs.

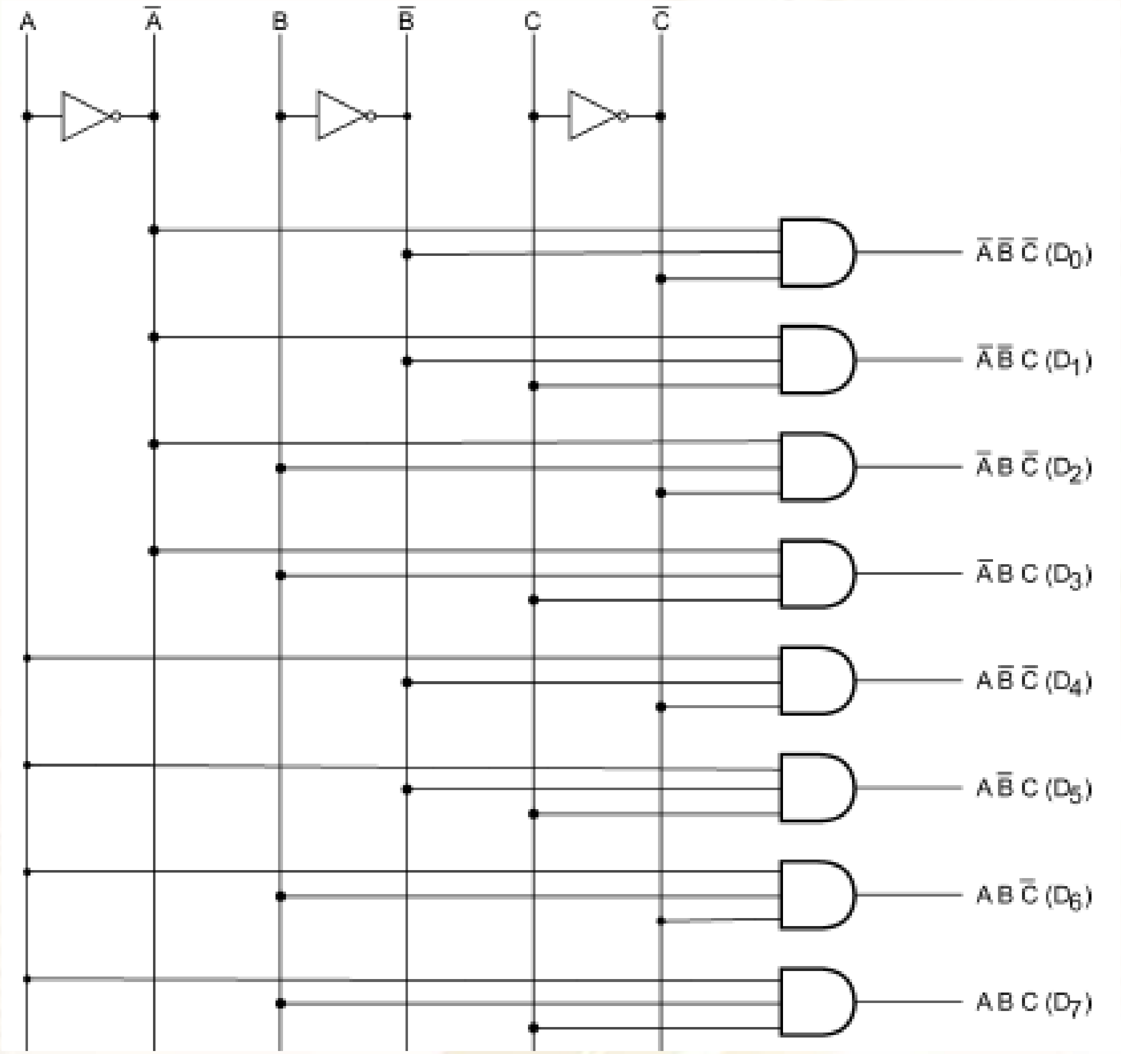
Is it an **active HIGH** or **active LOW** circuit? **Active HIGH**

3-line-to-8-line Decoder – Truth Table

- It has **3 inputs** and **8 outputs**
- **Truth Table** is given below:

INPUTS			OUTPUTS							
A	B	C	D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7
0	0	0	1	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0	0	0	0
0	1	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	1	0	0	0	0
1	0	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0	1	0	0
1	1	0	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	1

3-line-to-8-line Decoder Implementation

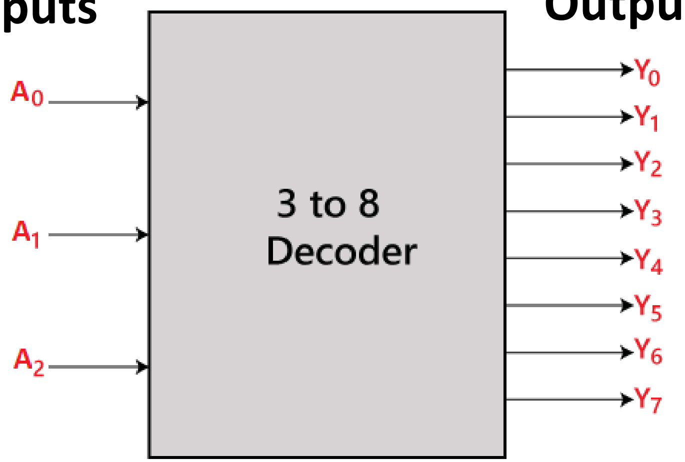


Logic Symbols of Decoders

1-of-8 or 3-to-8 Decoder
(active-HIGH outputs)

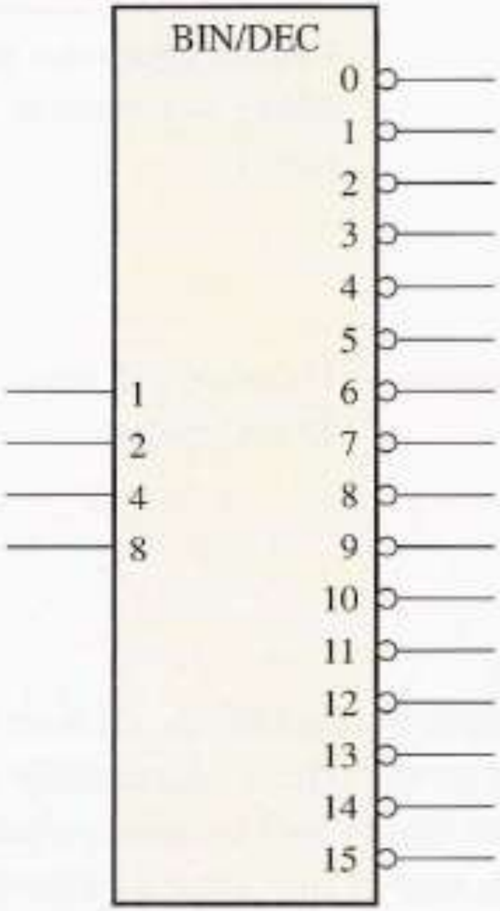
3-bit
Inputs

8-bit
Outputs



1-of-16 or 4-to-16 decoder
(active-LOW outputs)

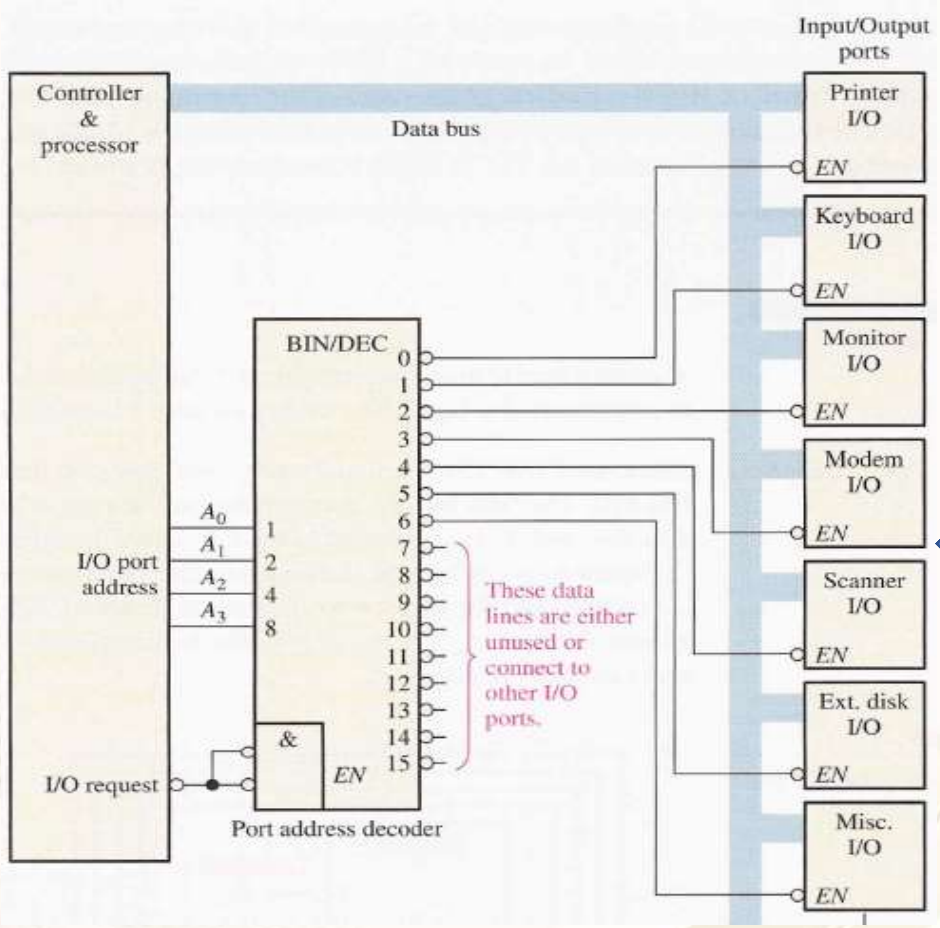
4-bit
Inputs



16-bit
Outputs

Application of Decoder

- An **I/O Port** address decoder



When will this **Modem I/O** module be **enabled**?
Or
What is the **I/O address** of this **Modem I/O** module?

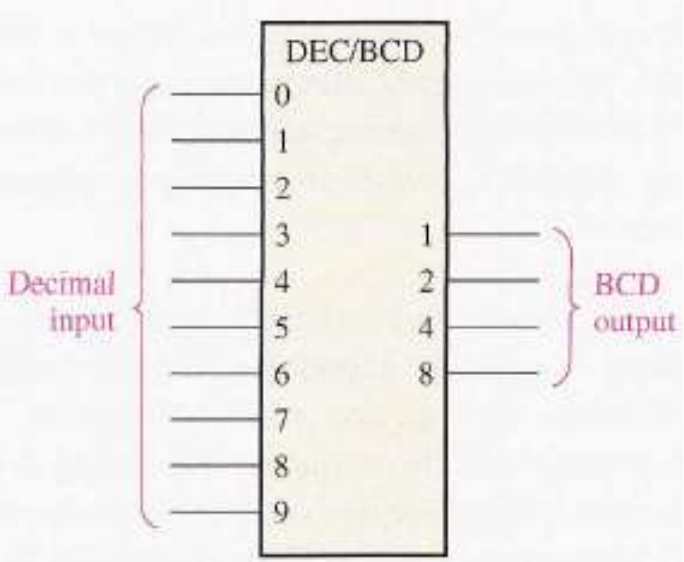
ANS: 0011 or 3

Encoder

Encoder

- An **encoder** is a combinational circuit that essentially performs a **reverse of a decoder** function
- An encoder accepts an active level on one of its inputs representing a digit, such as a decimal, or octal digit, and converts it to a coded output, such as BCD or binary

Decimal-to-BCD Encoder



DECIMAL DIGIT	BCD CODE A_3	A_2	A_1	A_0
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1

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Session 2.8 : Multiplexers

Session 2.8: Focus

- Multiplexers (MUX)
  - 2-to-1 line MUX Implementation
  - 4-to-1 line MUX Implementation
  - MUX Symbols/Representation
- Multiplexers (MUX)
- Real-life Applications of MUX

Multiplexer (MUX)

Multiplexer (MUX)

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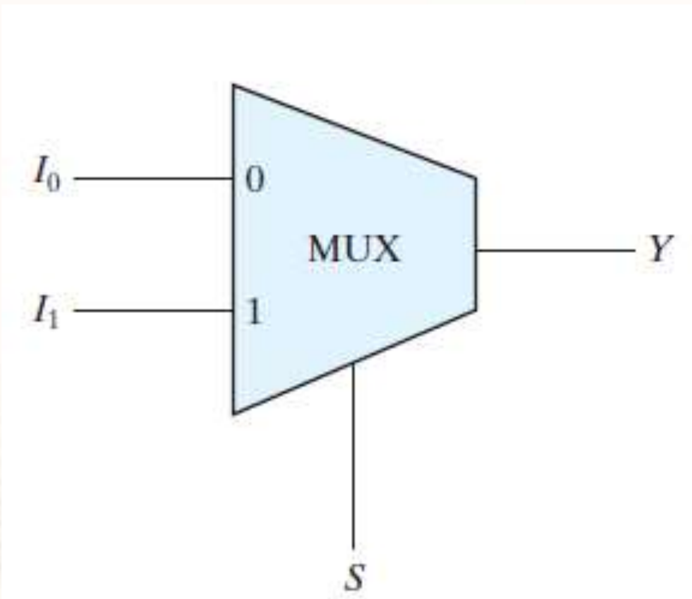
- A multiplexer (**MUX**) is a device that **allows** digital **information** from **several sources** to be **routed** onto **a single line** for transmission
- A basic **MUX** has **several data-input lines** and **a single output line**
- It has **data-select inputs**, which permit digital data on any one of the **inputs** to be **switched** to the **output** line
- Multiplexers are also known as **data selectors**

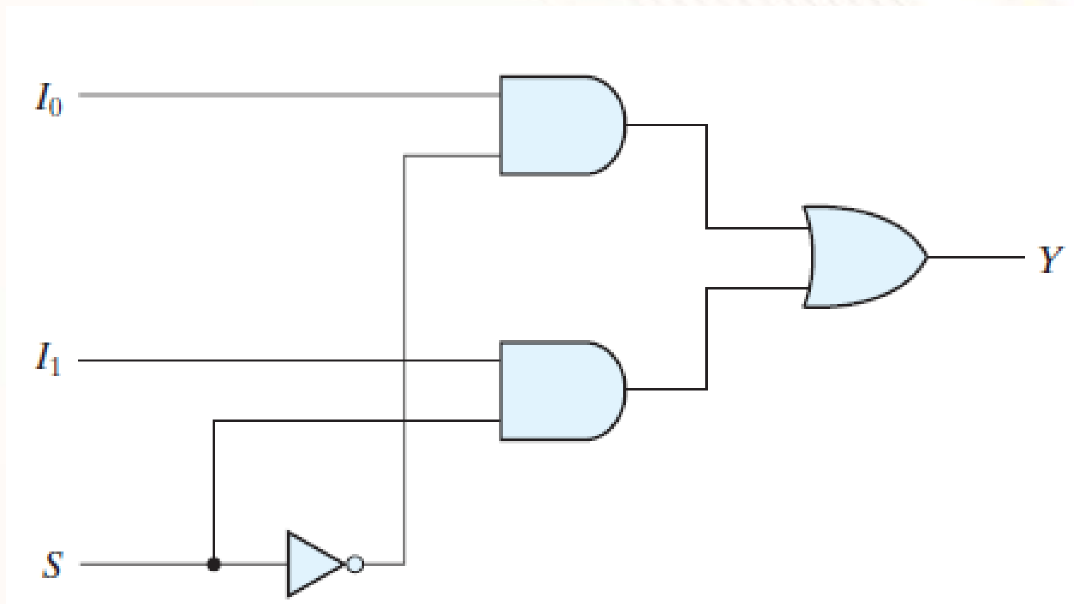
2-to-1 line Multiplexer

- When **S = 0, I_0** will be available at **Y**
- When **S = 1, I_1** will be available at **Y**

S is a **Select** Signal



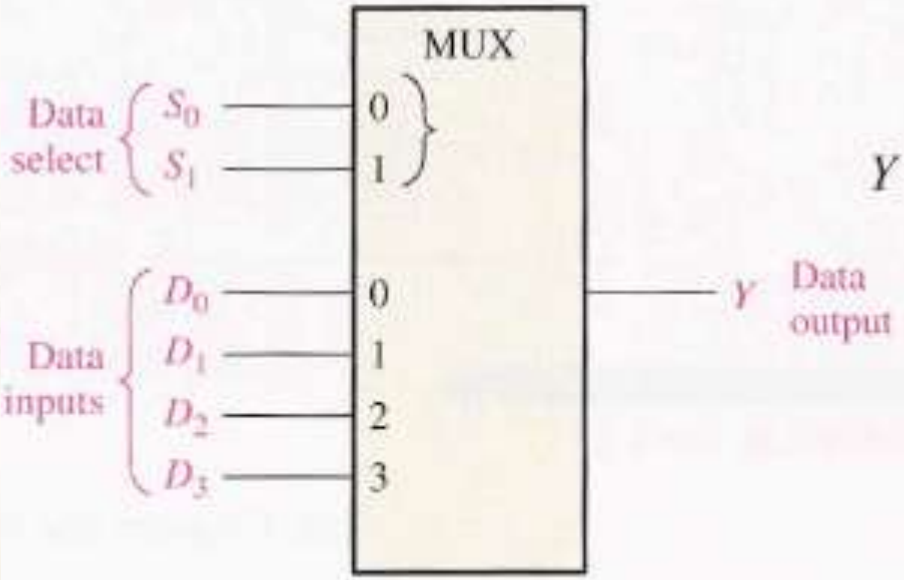
(b) Block diagram



(a) Logic diagram

4-to-1 line Multiplexer

DATA-SELECT INPUTS S_1	S_0	INPUT SELECTED
0	0	D_0
0	1	D_1
1	0	D_2
1	1	D_3



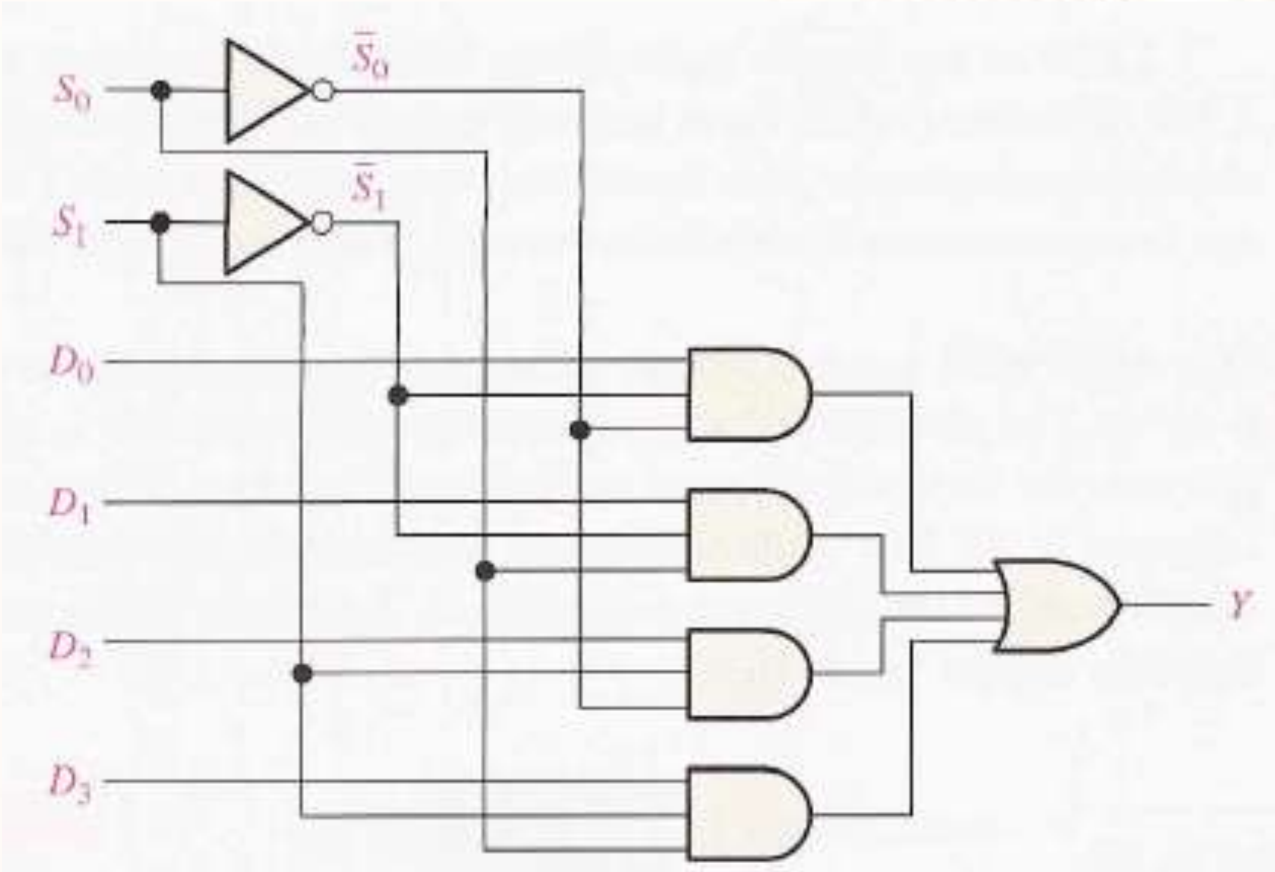
$$Y = D_0\bar{S}_1\bar{S}_0 + D_1\bar{S}_1S_0 + D_2S_1\bar{S}_0 + D_3S_1S_0$$

Implementation: 4-to-1 line Multiplexer

S_1	S_0	Y
0	0	D_0
0	1	D_1
1	0	D_2
1	1	D_3

(b) Function table

$$Y = D_0\bar{S}_1\bar{S}_0 + D_1\bar{S}_1S_0 + D_2S_1\bar{S}_0 + D_3S_1S_0$$

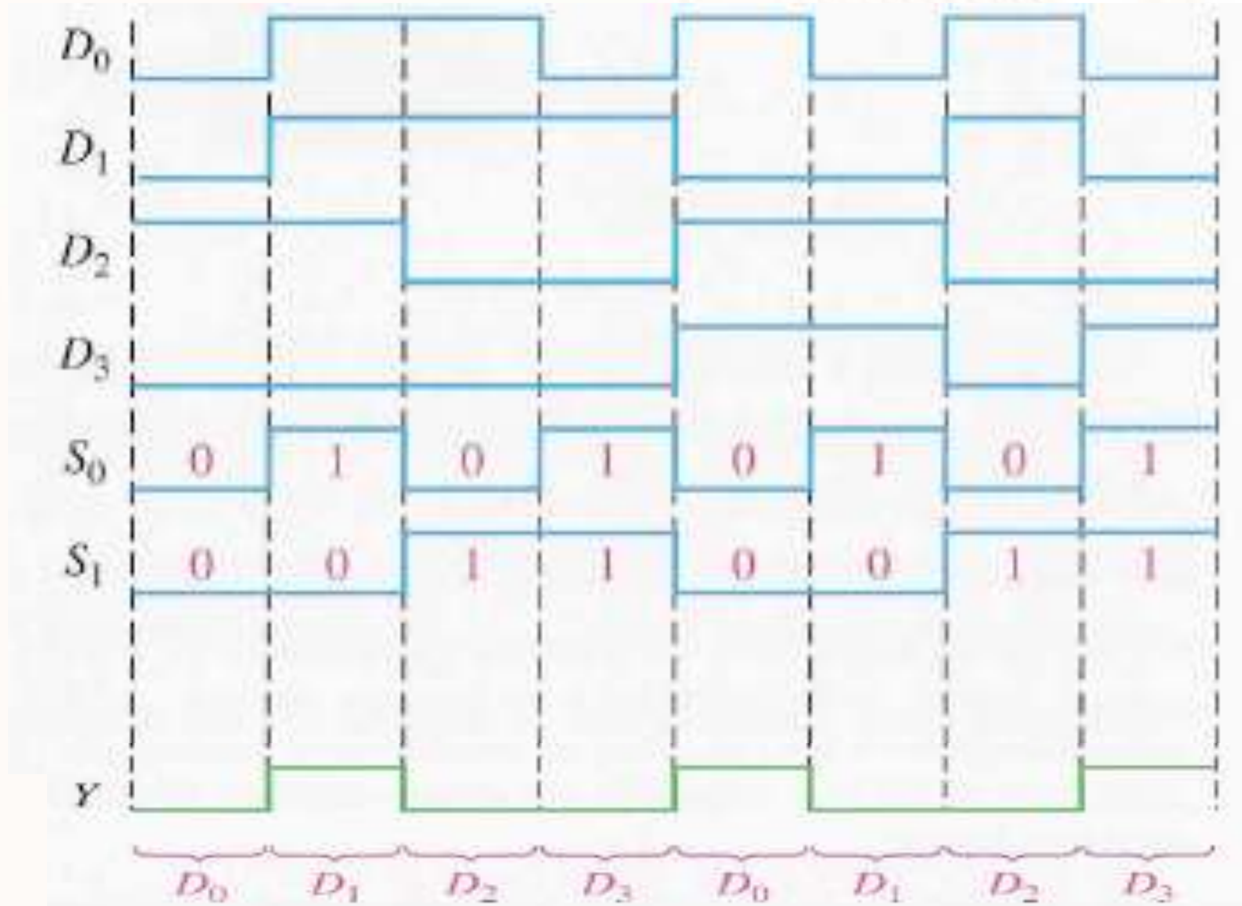


Quiz 1: Draw the Output Waveform

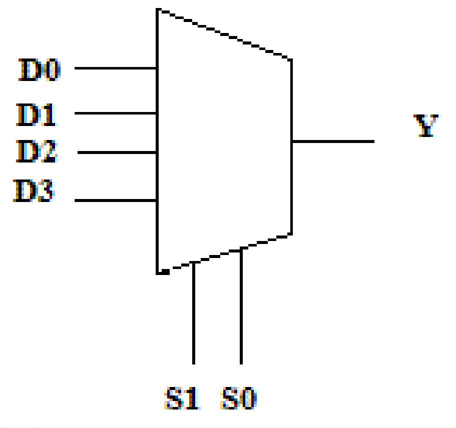
- Given the **data-input** and the **data-select waveforms**, **draw** the **output waveform**

S_1	S_0	Y
0	0	D_0
0	1	D_1
1	0	D_2
1	1	D_3

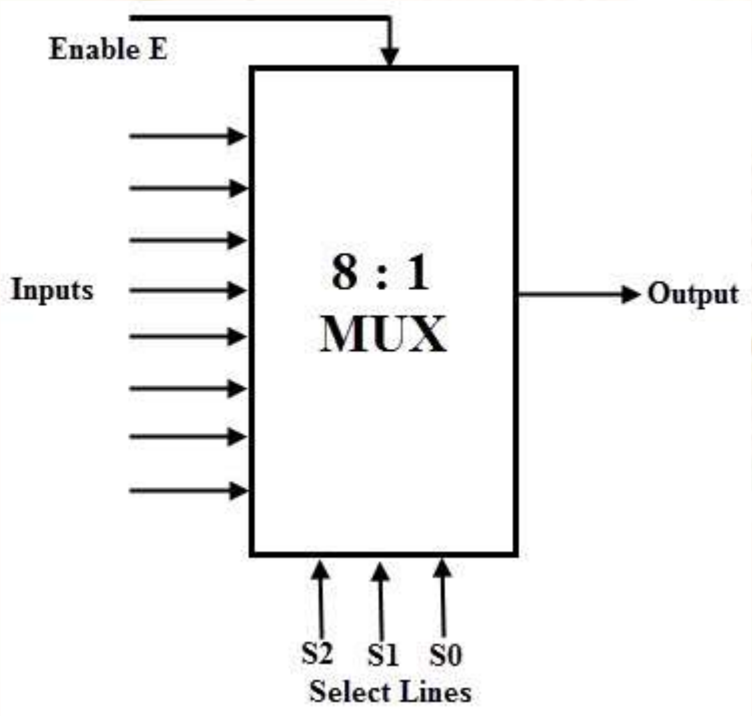
(b) Function table



Mux Symbols in Use

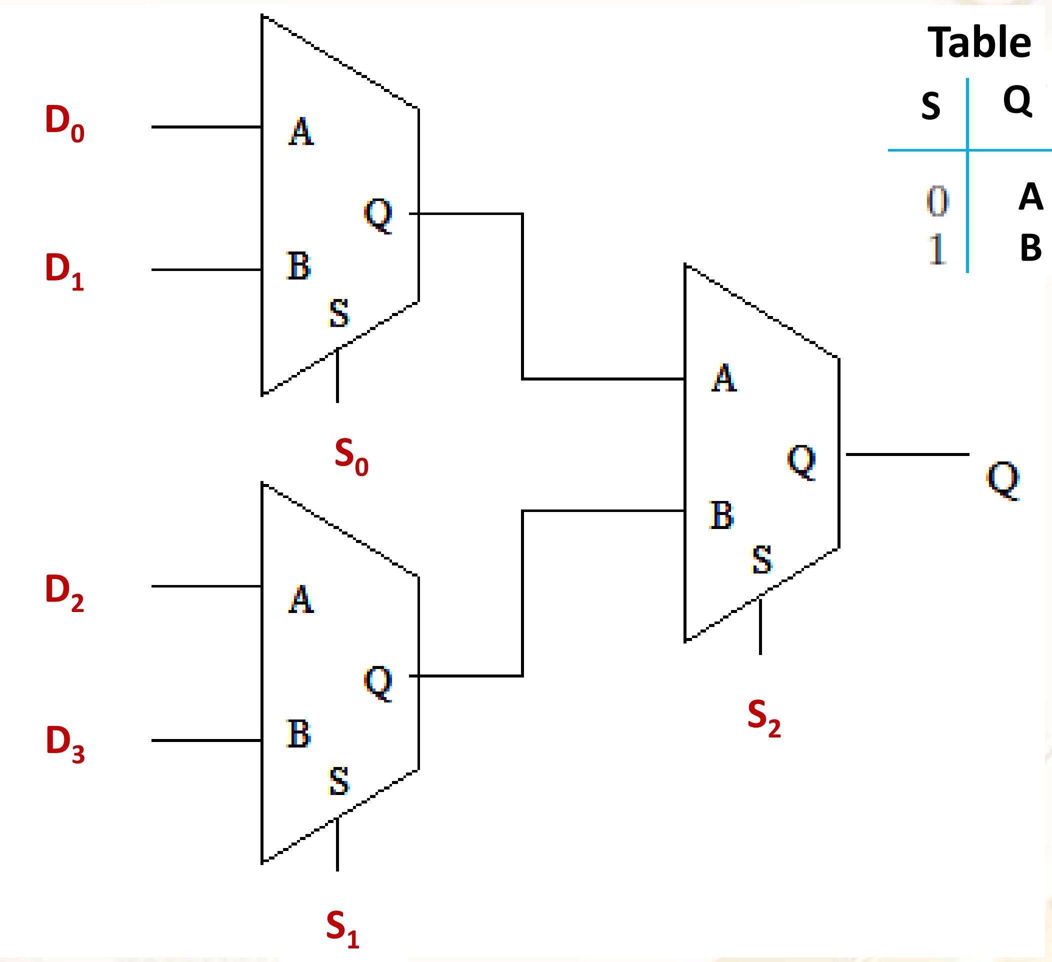


4-to-1 Line Mux



8-to-1 Line Mux

Quiz 2: What are the values at the output (Q)?



Function Table

S	Q
0	A
1	B

Selectors S_2	S_1	S_0	Output (Q)
0	0	0	D_0
1	1	1	D_3
1	0	1	D_2
0	0	1	D_1

Real-life Applications of MUX

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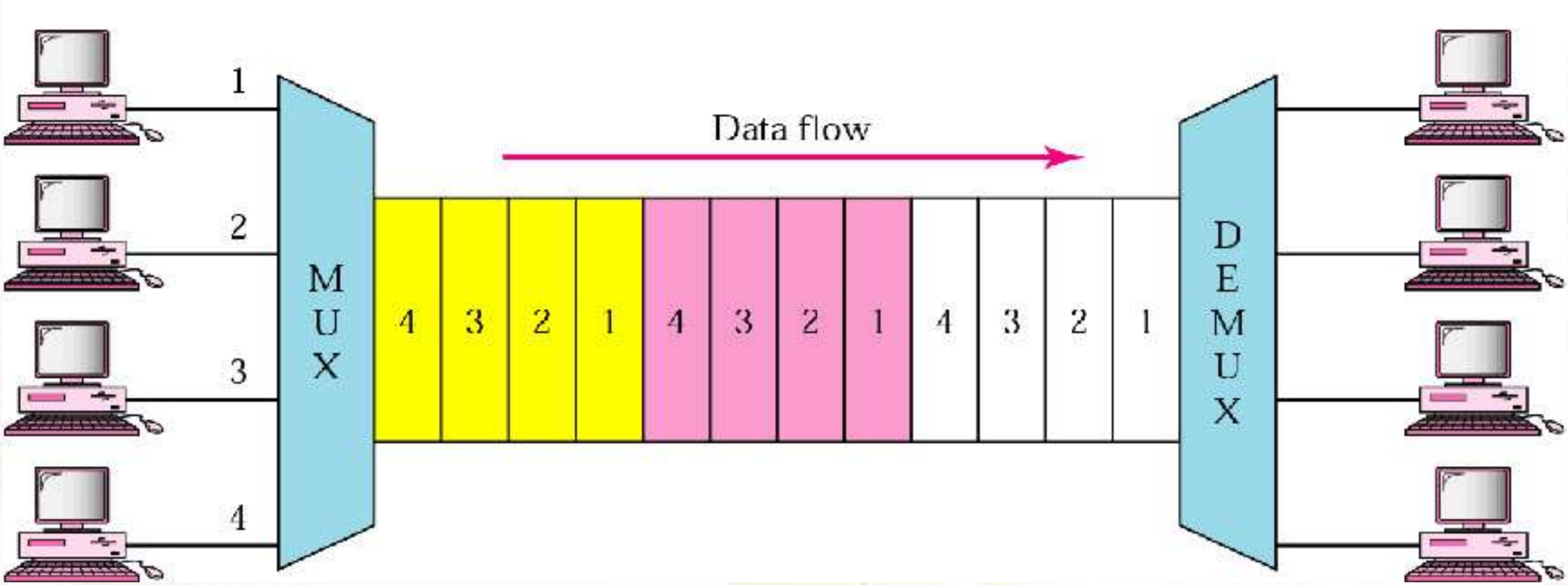
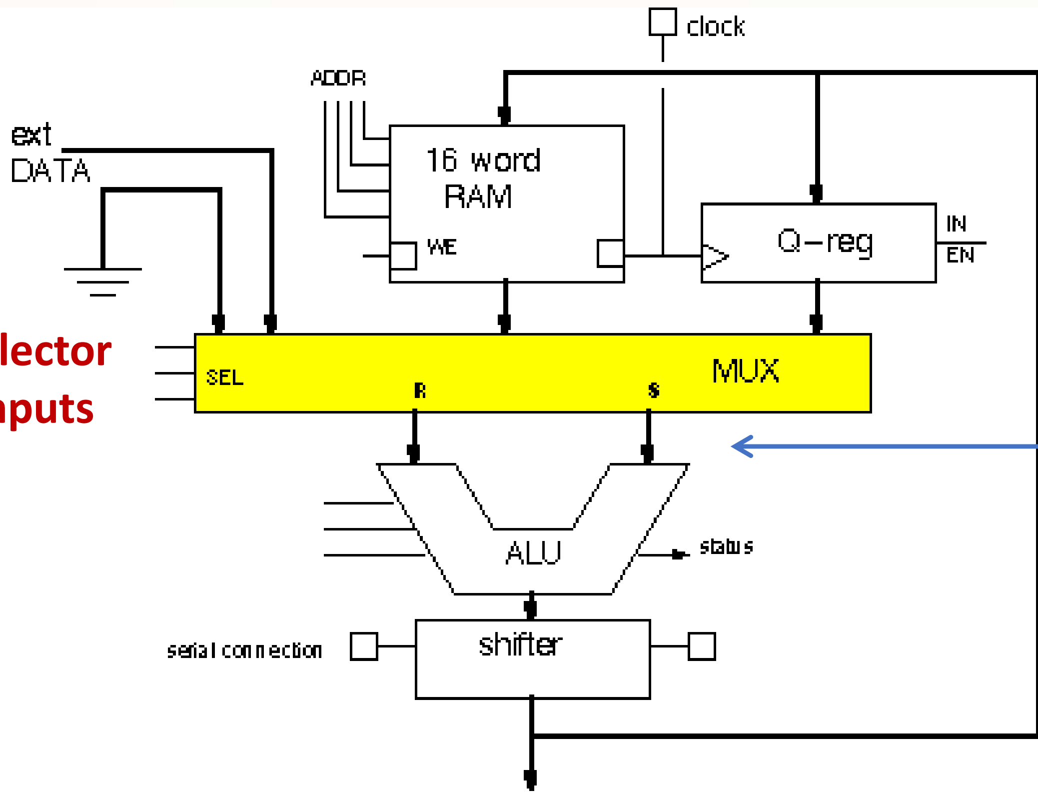
Multiplexer: Real-life Applications - 1

- **Time Division Multiplexer** (**TDM**) is one of the types of multiplexers which join data streams by allotting every stream a different time slot, in a sequence.



Multiplexer: Real-life Applications - 2

- Choosing one input from multiple input lines, to be given to Arithmetic Logic Unit (**ALU**)



Selector Inputs

Output Lines Bringing out One of the Input values

Session 2.8: Summary

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  - 2-to-1 line MUX Implementation
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- Multiplexers (MUX)
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Session 2.9 : Demultiplexers

Session 2.9: Focus

- Demultiplexer (DEMUX)
  - 1-to-4-line Demultiplexer
  - Symbols or Representation
  - Usage in Real-life Applications

Demultiplexer (DEMUX)

Demultiplexer (DEMUX)

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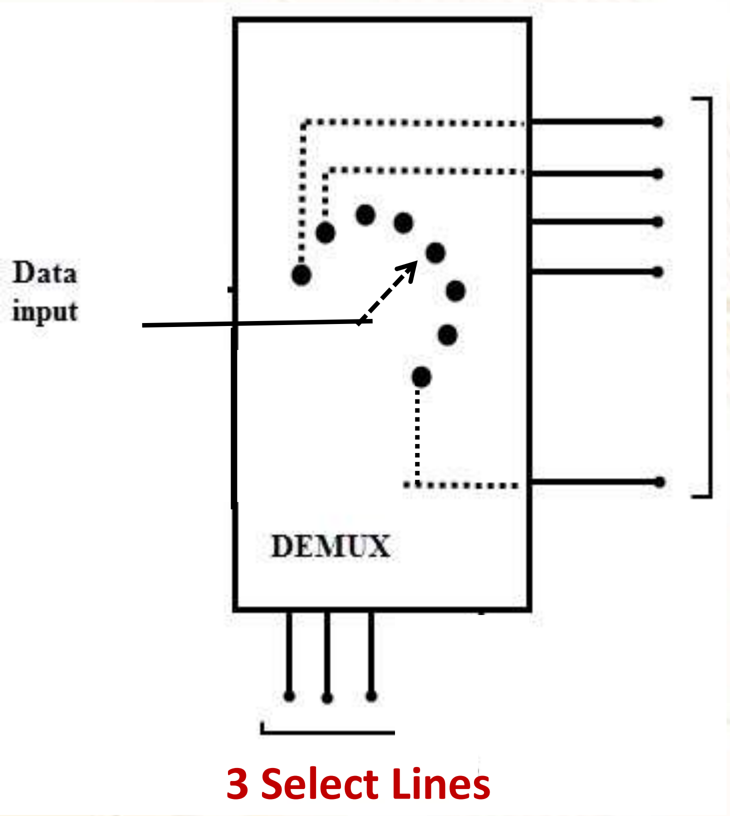
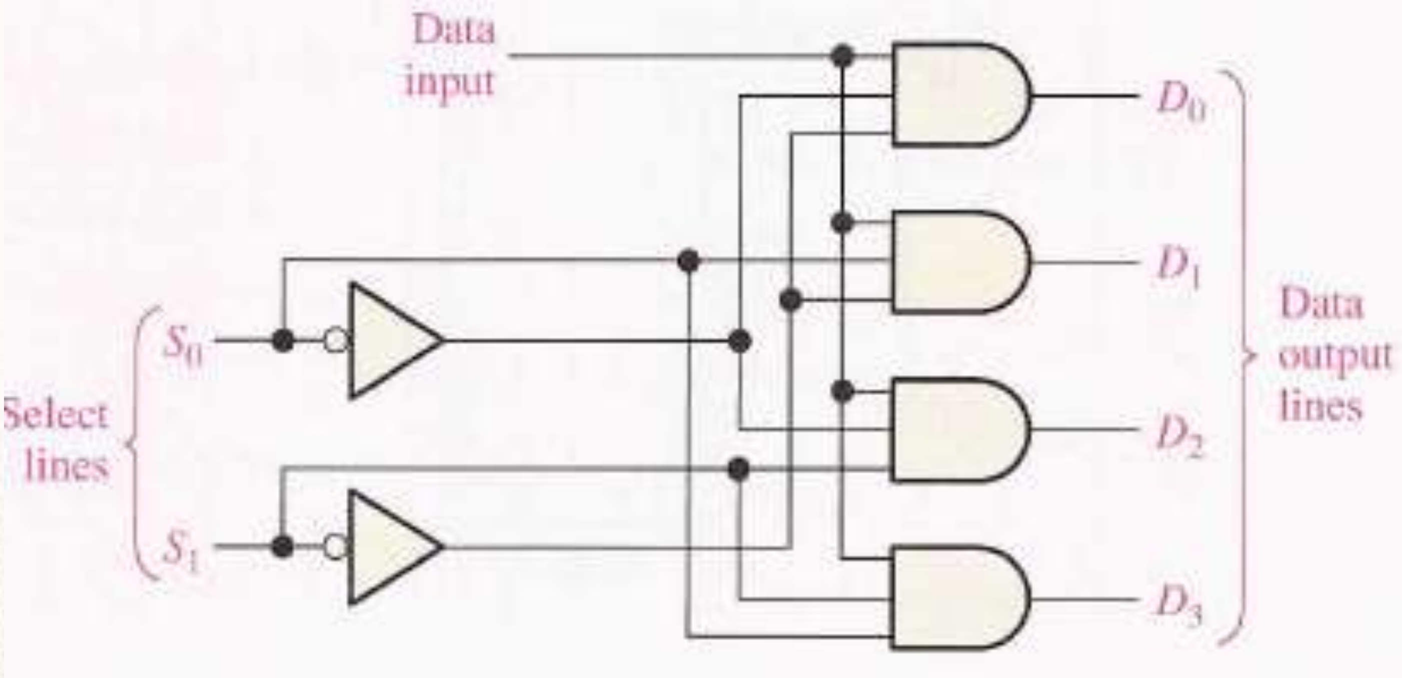
- A demultiplexer (**DEMUX**) basically reverses the multiplexing function
- It **takes** digital **information** from **one line** and **distributes** it to one **of the output lines**
  - Based on the **data-select** inputs
- For this reason, it is also known as **data distributor**



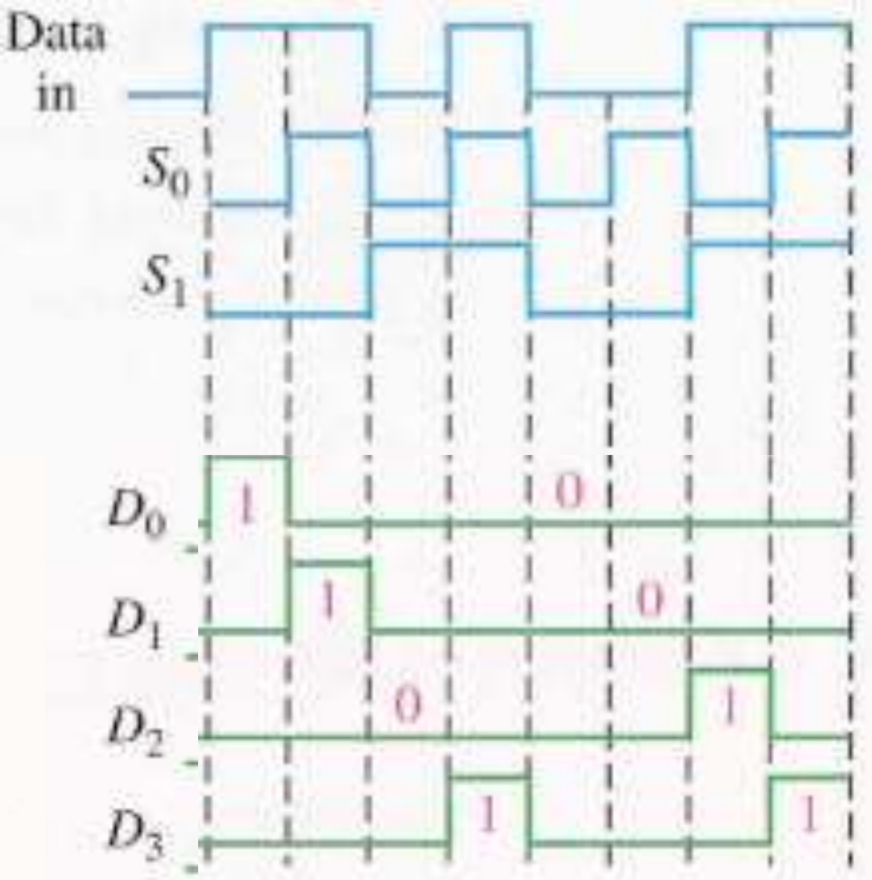
1-to-4 line Demultiplexer

- For example, if the **data select** lines are (**S_1S_0 - 10_b**) then **data input** will be sent out through **D_2** output line

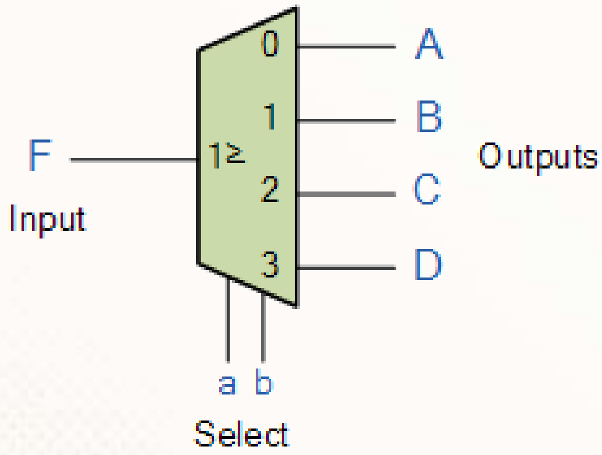


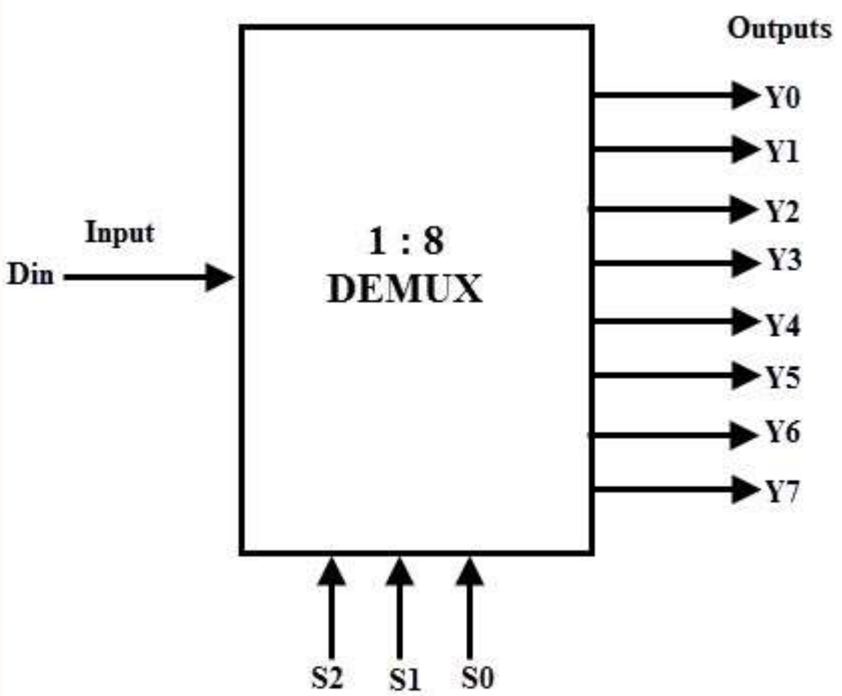
Quiz 1: Draw the Output Waveforms

- Given **data-input** and **data-select waveforms**, **draw** the **output waveforms on D_0 to D_3**

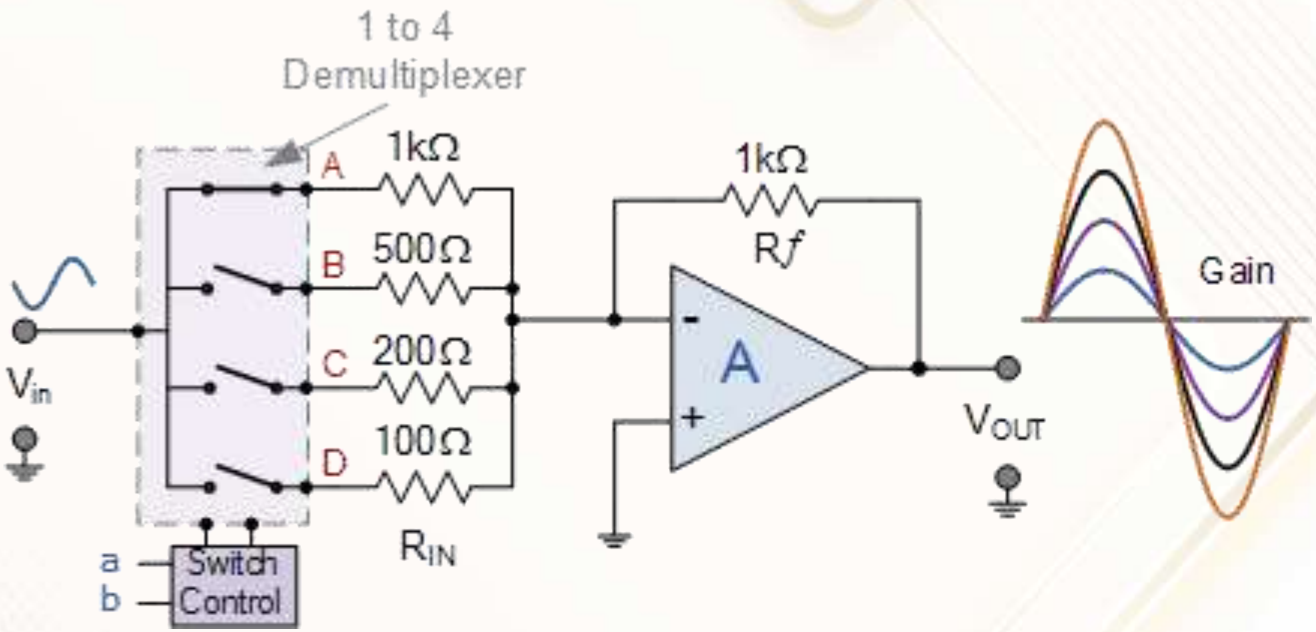


DEMUX: Symbol or Representations





Demux : Application



One of the Paths (A/B/C/D) is chosen,
to get different amplification of input signal

A: OpAmp: Operational Amplifier

Session 2.9: Summary

- Demultiplexer (DEMUX)
  - 1-to-4-line Demultiplexer
  - Symbols or Representation
  - Usage in Real-life Applications